EIE2111
Computer Programming

Dr. Lawrence Cheung

Mr. Richard Pang

Semester 1, 2021/22

Objectives

1. To introduce the fundamental concepts of computer programming
2. To equip students with sound skills in C/C++/C# programming language
3. To equip students with techniques for developing structured computer programs
4. To demonstrate the techniques for implementing engineering applications using computer programs

Objectives

Mission

You can write a C++ program by yourself!

- Do not worry, the course is designed for beginners.
  - But, of course, you will have the advantage if you learned computer programming before.

Methodology

- To write a computer program, you need to have
  - ***Fundamental concepts, programming syntax*** (I will teach)
  - ***Logical thinking*** (I cannot teach but you can learn it from laboratory exercises)
  - ***Debugging*** (find out errors from a program and fix them) (I cannot teach but you can learn it from laboratory exercises)

Teaching Staff

- Lecturer: Dr. Lawrence Cheung
  - Teach Chapter 1 to 5
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Teaching Staff

- Lecturer: Mr. Richard Pang
  - Teach Chapter 6 to 8
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Classroom Discipline

Principle 1:

Do not disturb my teaching

Principle 2:

Do not disturb your classmates

if they are listening to me

It is not important in online mode now!

Classroom Discipline

Therefore …

- Please ask me questions during my teaching. It is important to let me know what you do not understand; otherwise, you will get lost in the rest of the lecture.
- You can type in your question in the chat room and I will reply once I find it.

Preparation

- Students should read through the lecture notes ***at least once*** before attending the class.
- Students are advised to ask specific questions in relation to specific topics rather than asking vague questions.
  - Example: Don’t ask “I don’t understand Chapter 3. Can you teach me again?”

Email Format

- Should there be any difficulties or problems during non-office hours, the best way to contact me is via email.
- Email format:

> **<u>Subject: EIE2111 - xxx</u>**
>
> Dear Dr. Cheung,
>
> I have a question about … … …
> Thanks for your attention.
>
> Regards,
>
> Chan Tai Man
> 1234567D

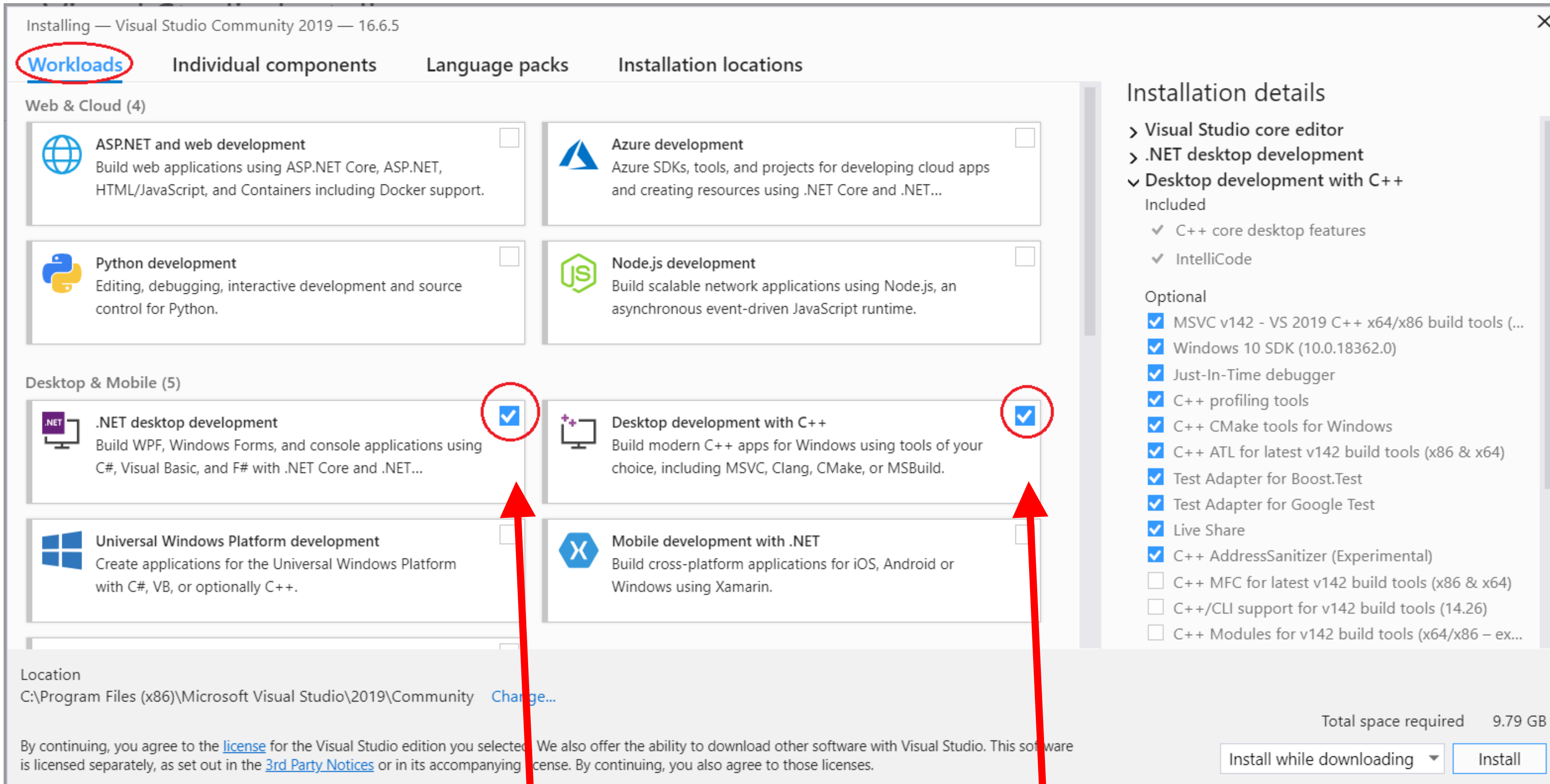
Useful Information

- Course web sites
  - Blackboard
- Textbook
  - H. M. Deitel and P. J. Deitel, "C++ How to Program", 10th edition, Prentice Hall, 2016.

Course Schedule

- Lecture
  - Thursday, 11:30 a.m. – 1:30 p.m., PQ303
- Laboratory
  - Tuesday, 11:30 a.m. – 1:30 p.m., CF502, CF503 and CF504
  - Start from Week 2

Visual Studio Community 2019



https://visualstudio.microsoft.com/

Use default setting to install C++ & C# core components

Course Outline (Semester 1)

Week	Lecture	Lab.
1	Course Information Introduction (Ch. 1) Introduction to C++ Programming (Ch. 2)	Nil
2	Introduction to Classes and Objects (Ch. 3)	Lab 1: Getting Started and Debugging (Ch. 1 and 2)
3	Control Statements: Part 1 (Ch. 4)	Lab 2: Classes and Objects (Ch. 3)
4	***<u>No lecture</u>***	Lab 2: Classes and Objects (Ch. 3)

Course Outline (Semester 1)

Week	Lecture	Lab.
5	Control Statements: Part 2 (Ch. 5)	Quiz 1 (Ch. 3) Lab 3: Control Statements (Part 1) (Ch. 4)
6	Functions and an Introduction to Recursion (Ch. 6)	Lab 3: Control Statements (Part 1) (Ch. 4)
7	***No lecture*** ***(Chung Yeung Festival)***	Quiz 2 (Ch. 4) Lab 4: Control Statements (Part 2) (Ch. 5)
8	***No lecture***	Lab 4: Control Statements (Part 2) (Ch. 5)

Course Outline (Semester 1)

Week	Lecture	Lab.
9	Functions and an Introduction to Recursion (Ch. 6) Arrays and Vectors (Ch. 7)	Quiz 3 (Ch. 5) Lab 5: Functions and Recursion (Ch. 6)
10	***No lecture***	Lab 5: Functions and Recursion (Ch. 6)
11	Arrays and Vectors (Ch. 7)	Quiz 4 (Ch. 6) Lab 6: Arrays and Vectors (Ch. 7)
12	***No lecture***	Lab 6: Arrays and Vectors (Ch. 7)
13	***No lecture***	Quiz 5 (Ch. 7)
14	Final Test 1	

Course Outline (Semester 2)

Week	Lecture (tentative)	Lab. (tentative)
1	Pointers, Command-line Processing and Linked Lists (Ch. 8, Appendix E and Ch. 21)	Lab 7: Command-line processing and Linked Lists
2	Pointers, Command-line Processing and Linked Lists (Ch. 8, Appendix E and Ch. 21)	
3	File Processing (Ch. 17)	Lab 8: File Processing Quiz 6 (Ch. 8, Appendix E and Ch. 21)
4	***No lecture***	

Course Outline (Semester 2)

<table>
<tr><th>Week</th><th>Lecture (tentative)</th><th>Lab. (tentative)</th></tr>
<tr><td>5 – 7</td><td><u>No lecture</u></td><td>Quiz 7 (Ch. 17, Week 5)
GUI (Lecture)
Lab 9: GUI</td></tr>
<tr><td>8 – 9</td><td colspan="2">Assignment 2
Quiz 8 (GUI, Week 8)</td></tr>
<tr><td>10 – 13</td><td colspan="2">Mini-Project</td></tr>
<tr><td>14</td><td colspan="2">Final Test 2</td></tr>
</table>

Continuous Assessment

- Laboratory Exercises (10%, 1%@)
  - Nine laboratory exercises, 1% each, except 2% for “Lab 2: Classes and Objects".
  - You will have time in laboratory sessions to do your laboratory exercises. But you should also do it at home.
  - They are very important. It is ***essential*** for you to practice your programming skill.

Continuous Assessment

- You are welcome to ask for help when you have difficulties to do the laboratory exercises.
- Zip all project files into one single file and submit it to ***Blackboard*** on or before the deadlines.
- You will score ***no marks*** if you submit after the deadline.

Continuous Assessment

- ***Deadlines***
  - Lab 1: 5 pm, 10 September 2021 (Friday, Week 2)
  - Lab 2: 5 pm, 24 September 2021 (Friday, Week 4)
  - Lab 3: 5 pm, 8 October 2021 (Friday, Week 6)
  - Lab 4: 5 pm, 22 October 2021 (Friday, Week 8)
  - Lab 5: 5 pm, 5 November 2021 (Friday, Week 10)
  - Lab 6: 5 pm, 19 November 2021 (Friday, Week 12)

Continuous Assessment

- Assignments (10%, 5%@)
  - Deadline of Assignment 1: ***<u>5 pm, 7 Jan 2022</u>***
    - Early bird deadline: ***<u>5 pm, 31 Dec 2021</u>***
    - Early bird bonus: ***<u>total marks x 1.05</u>***
  - Assignment 2 should be submitted in the middle of Semester 2.
  - You are required to submit your assignments to Blackboard on or before the deadlines.

Continuous Assessment

- You will score ***no marks*** if you submit after the deadline.
- You may score ***Grade F*** in this subject if any cheating is found in your assignments.
- You ***must*** get my approval ***before*** the deadline if you look for the late submission. You may not get the approval if you do not have strong reasons (e.g., sick with medical certificates).

Continuous Assessment

- ***Do not copy assignments from your classmates***. I have a software to check the similarity. It worked very well before.
- After grading, you can see your results through Blackboard. You are welcome to discuss your performance with me, if any.

Continuous Assessment

- Mini-Project (30%)
  - Design and implement a small computer game.
  - It starts from the middle of Semester 2 and should be finished at the end of Semester 2.
  - You are required to submit your mini-project to Blackboard on or before the deadline.

Continuous Assessment

– ***Do not copy the mini-project from your classmates.***
– You may score ***Grade F*** if any cheating is found in your mini-project.

Continuous Assessment

- 8 Quizzes (8%, 1%@, 1 hour@)
  - 11:45 a.m. – 12:45 p.m.
  - Open-book practical test
    - You are allowed to bring your lecture notes, books, and any secondary storages.
    - You may use a PC or your notebook in our labs to write your program.
    - You are NOT allowed to get any information from the Internet. NO web browser.
  - ONE programming question (two versions)

Continuous Assessment

- At the end, you should zip your project files into one single file and submit it to Blackboard.
- Then you demonstrate your program to show whether your program works or not.
- If you do not know why your program does not work in a quiz, we will come to help you after demonstration.

Continuous Assessment

If you cannot finish a quiz properly, you have to work it out and submit to Blackboard before you attend the next quiz; otherwise, you cannot attend the next quiz.

We will check.

Continuous Assessment

- 2 Final Tests (42%, 21%@)
  - Close-book written test (3%)
    - TWO to THREE short questions
  - Open-book practical test (18%)
    - TWO programming questions (two versions)
    - At the end, you should zip your project files into one single file and submit it to Blackboard.
    - Then you demonstrate your program to show whether your program works or not.

Continuous Assessment

- Marking Criteria for Quizzes and Final Tests:
  1. You will score NO marks in a programming question if your program cannot be compiled successfully.
  2. You will score NO marks in a programming question if your program cannot show any outputs properly.
  3. You will score some marks in a programming question if your program can show some reasonable outputs.
  4. You will score FULL marks in a programming question if your program can give the expected outputs in our testing cases.
  5. You are welcome to discuss your case with me if you confuse about my marking.

Important Issues

- Quizzes and Final Tests
  - To avoid cheating, you should not open any applications except PDF reader and Microsoft Visual Studio 2019.
  - You should not open any web browsers.
  - All flash drives must be removed before the test starts; otherwise, you will score ***<u>no marks</u>*** in the test.

Final Test 1

- Date: ***29 November 2021*** (Monday)
- Time: 9:30 a.m. – 1:00 p.m.
  - Close-book written test: 10 minutes, 3%
  - Open-book practical test: 170 minutes, 18%
- Write it down. It is confirmed.
- Venue: to be confirmed

Methodology (re-visit)

- Fundamental concepts, programming syntax
  - Teach in lecture session.
  - Some simple class exercises will be provided in lectures.
- Logical thinking
  - Learn from laboratory exercises.
  - ***Ask for help*** if you have any difficulties. It is very important.

Methodology (re-visit)

- Debugging
  - Learn from laboratory exercises.
  - Thus, it is very important.
  - You will learn ***nothing*** if you simply get the answers from your classmates.
  - You will almost learn ***nothing*** if you do not work out your idea in programming.

Advices

1. **The best way to learn computer programming is practice.**

2. **Please ask for help when you have problems.**

3 **You will learn nothing if you do not work out your program by yourself.**

Advices

About Point 3:

"I try but I fail. Then I ask my classmates for help. They tell me the idea and I finally work out the program."

"I try but I fail. I ask Lawrence for help. He helps me to make my idea works. Thanks."

Advices

- That is why
  - You will have quite ***a lot of lab. exercises*** for you to practice your programming skill.
  - You will spend quite ***a lot of time*** (two hours per week) to do the lab. exercises in the class.
  - You will spend quite ***a lot of time*** to do the lab. exercises at home. (you can install Microsoft Visual Studio 2019 in your home computer or notebook)

Advices

– Some concepts you may not understand very well after attending lectures. But, you may find that it is much better after you finish the lab. exercises.

Advices

- Debugging (find out errors from a program and fix them) is very important.
  - ***Always read*** the ***first*** error message, understand it, and fix it. Don't skip the first one.
  - All error messages are written in ***English***, you can understand them if you read them carefully.
  - No one will help you in quizzes and tests.

Advices

- All examples in the lecture notes are very important.
  - Their program codes can be found in the course web site.
  - They are very useful. You should make use of them when you do your lab. exercises, assignments, mini-project, quizzes, and tests.
- Save all your works and make more than one copy. You may use them later.

End